

WHEN FUZZINESS IS IN THE MIND, NOT IN THE CONCEPT: TERMINOLOGY, PROTOTYPES, AND ONTOLOGICAL MODELING

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ABSTRACT. In recent decades, terminology studies have increasingly engaged with findings from cognitive psychology, corpus linguistics, and discourse analysis in order to account for variation in term usage, historical change, and context-dependent meaning (Cabr , 1999; Temmerman, 2000). This development has led to the emergence of socio-cognitive approaches to terminology, which challenge classical W sterian models of concepts traditionally associated with standardization-oriented terminology work. In particular, prototype-based theories of categorization have been mobilized to argue that many concepts lack clear boundaries and that classical Aristotelian definitions based on necessary and sufficient characteristics are inadequate for describing real-world terminological practice (Lakoff, 1990; Rosch, 1975, 1973). These socio-cognitive insights have contributed significantly to the descriptive analysis of terminological variation and to the discussion of the possible pitfalls of the general theory of terminology. Nevertheless, this paper argues that the theoretical implications of socio-cognitive approaches for concept modeling, ontology design, and terminology management have often been overstated. The central claim we try to defend here is that graded categorization behavior and prototype effects pertain to cognitive access and usage patterns, not to the ontological structure of concepts themselves. Failure to maintain this distinction has often led to an unnecessary dichotomy (classical vs modern terminology, right vs wrong, etc.) with a consequent conceptual confusion and practical difficulties in applied terminology and ontology engineering.

1. INTRODUCTION

From an applied perspective, terminology work operates under constraints that require conceptual precision, stability, and interoperability, particularly in domains such as science, technology, medicine, engineering, and any other domain for

Preprint version. Accepted for presentation at TOTH 2026, the 20th anniversary conference. Conference website: <https://toth.condillac.org/>.

which we (researchers) experience a high degree of change and innovation. Terminology resources are increasingly integrated into digital infrastructures, including databases, knowledge graphs, semantic web applications, and AI-driven decision-support systems (Di Nunzio, 2025; Gruber, 1993; Guarino et al., 2009; Khemakhem et al., 2025). In these contexts, concepts must be modeled in a way that supports logical inference, data integration, and consistent interpretation across systems and communities. This data modeling approach where concepts in an ontology (in a computer science sense) exist and are distinguishable one from the other is almost perfectly matched by the classical view of terminology, particularly in the tradition associated with Eugen Wüster. This approach, in fact, sees concepts as units of knowledge defined by characteristic sets that allow them to be clearly distinguished within a concept system. From an applied standpoint, its strength lies precisely in its normative orientation: it provides principles for constructing terminology resources that are operational, reusable, and aligned with domain knowledge rather than with individual perspectives of what a concept could be. Importantly, classical terminology does not deny that users may experience difficulty in applying concepts; rather, it treats such difficulty as a matter of education, expertise, or interface design, not as evidence of conceptual indeterminacy. More recently (Di Nunzio et al., 2025), this distinction has been explicitly addressed in applied work showing how disagreements among domain experts about the definition or scope of a concept often reflect differences in perspective, interpretation, or contextual emphasis rather than genuine indeterminacy at the conceptual level. In such cases, the observed “fuzziness” arises from the interaction between experts and the concept, shaped by disciplinary background, task-specific goals, or evidential priorities, rather than from the absence of a determinate underlying concept. From this viewpoint, divergent expert definitions do not necessarily call for weakening conceptual boundaries in terminological or ontological models; instead, they motivate the explicit representation of alternative viewpoints. Socio-cognitive terminology, most prominently articulated by Temmerman (2000), emphasizes that many terminological categories, especially in the life sciences and other rapidly evolving domains, do not exhibit the stability presupposed by classical definitions. Drawing on prototype theory and discourse analysis (Rosch, 1975), Temmerman highlights the role of typicality, metaphor, historical development, and community-specific practices in shaping terminological meaning. These observations are particularly relevant for applied terminology management, where terminologists must often deal with competing definitions, overlapping concepts, and evolving domain knowledge. In this sense, on the one hand it is true that the classical approach is often criticized as overly rigid or disconnected from actual language use. On the other hand, it is also true that treating concepts as inherently fuzzy or prototype-structured at the ontological level poses serious challenges for applications that use knowledge bases and ontologies for mapping terminological data. This has often resulted in an unnecessary opposition between classical and socio-cognitive

terminology, framed as a choice between rigid formalism and descriptive adequacy, rather than as a difference in explanatory level.

2. OUR PROPOSAL

With the same spirit of our previous works dedicated to finding positive similarities among different approaches which would help (at least in theory) to allow for integration of different resources, in this paper we argue that socio-cognitive terminology tends to conflate three distinct levels of analysis that should be kept separate for both theoretical clarity and practical effectiveness:

- First, the ontological level of concepts as elements of a domain,
- Second, the definitional level of terminological descriptions and standards, and
- Third, the cognitive level of how users access, interpret, and apply concepts in practice.

Prototype effects and graded typicality belong primarily to the third level. When they are projected onto the first level, they introduce unnecessary vagueness into conceptual models that are meant to support precise reasoning and interoperability. The practical consequences of this conflation become especially evident when terminology is integrated with ontology engineering and artificial intelligence (Smith, 2001). In AI and knowledge representation, a clear distinction is maintained between ontologies, which provide formal, explicit specifications of concepts and relations, and user models, which capture how agents reason about, misunderstand, or prioritize information. Ontologies are evaluated with respect to consistency, coverage, and inferential adequacy, while user models are designed to accommodate variability, uncertainty, and heuristic reasoning. Importantly, inaccuracies or hesitations in user classification do not motivate changes to the ontology itself; instead, they motivate improvements in user support, explanation facilities, and interface design. This separation offers a powerful applied framework for terminology studies. Concepts can be modeled ontologically in a determinate and theory-driven manner, while prototype effects can be addressed through annotations, usage notes, corpus-based evidence, and adaptive user interfaces. For example, terminological databases can maintain precise concept definitions while also recording typical examples, frequency patterns, or domain-specific salience information to support users with different backgrounds and levels of expertise. Such an approach preserves the operational strengths of classical terminology while incorporating the descriptive insights emphasized by socio-cognitive approaches. From this applied standpoint, the often-cited opposition between Aristotelian and prototype-based views appears less as a fundamental theoretical conflict and more as a misunderstanding of explanatory scope. Prototype theory does not demonstrate that concepts are inherently fuzzy; it demonstrates that human interaction with concepts is mediated by cognitive heuristics (Fodor, 1998; Murphy, 2002;

Rosch, 1973). When this insight is properly situated at the level of cognitive access rather than concept structure, it becomes compatible with classical terminology and with contemporary ontology engineering practices. The paper therefore proposes a layered model for applied terminology and ontology work, consisting of:

- (1) Ontologically grounded concepts defined within a domain theory;
- (2) Terminological definitions that normatively describe these concepts for communication and standardization as well as empirical patterns of term usage observed in corpora and discourse;
- (3) Cognitive prototypes that influence how users process terms and apply concepts.

Each layer serves a distinct function and should be modeled using appropriate methods and tools. Adopting this layered approach has concrete benefits for applied terminology and AI-enabled systems. It supports the development of interoperable terminology resources, facilitates integration with formal ontologies and knowledge graphs, and enables the design of user-centered applications without compromising conceptual rigor. It also provides a principled way to incorporate socio-cognitive insights into terminology management while avoiding the erosion of conceptual clarity that is essential for standardization and automated reasoning.

3. FINAL CONSIDERATIONS

In this paper, we argue that the question is not whether terminology should account for cognitive and discursive variation, it should, but how this variation should be represented within terminological and ontological systems. By restoring a clear distinction between user-related cognitive processes and the formal structure of concepts, and by aligning terminology theory with established practices in ontology engineering and AI, this approach avoids conceptual confusion while preserving practical effectiveness. We discuss an applied and methodologically coherent framework for addressing fuzziness, variation, and typicality in contemporary terminology work. Importantly, this position does not deny the usefulness of representing uncertainty or vagueness in applied systems; rather, it aligns with existing work in ontology engineering that treats fuzziness as a modeling or reasoning layer, not as a property of concepts themselves.

DECLARATION ON GENERATIVE AI

During the preparation of this work, the author(s) used ChatGPT-5.4 in order to: Grammar and spelling check, Paraphrase and reword. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the publication's content.

REFERENCES

- Teresa Cabré. *Terminology. Theory, Methods and Applications*. John Benjamins, 1999. ISBN 978-90-272-9865-2. doi: 10.1075/tlrp.1.
- Giorgio Maria Di Nunzio. Terminology-Augmented Generation (TAG): Foundations, Use Cases, and Evaluation Paths. *Journal of Digital Terminology and Lexicography*, 1(1):97–104, July 2025. ISSN infase di registrazione. doi: 10.25430/pupj.jdtl.1752566034.
- Giorgio Maria Di Nunzio, Lucia Manni, Emanuela De Lisa, and Federica Vezzani. The Elephant in the Room: The Impact of Domain-Expert (Dis)agreement. A Case Study on Cryptic Species. In Federica Vezzani, Giorgio Maria Di Nunzio, Elpida Loupaki, Georgios Meditskos, and Maria Papoutsoglou, editors, *Proceedings of the 4rd International Conference on Multilingual Digital Terminology Today (MDTT 2025)*, volume 3990 of *CEUR Workshop Proceedings*, Thessaloniki, Greece, June 2025. CEUR. URL <https://ceur-ws.org/Vol-3990/\#short25>.
- Jerry A. Fodor. *Concepts: Where Cognitive Science Went Wrong*. Clarendon Press, February 1998. ISBN 978-0-19-151906-2.
- Thomas R. Gruber. A translation approach to portable ontology specifications. *Knowledge Acquisition*, 5(2):199–220, June 1993. ISSN 1042-8143. doi: 10.1006/knac.1993.1008.
- Nicola Guarino, Daniel Oberle, and Steffen Staab. What Is an Ontology? In Steffen Staab and Rudi Studer, editors, *Handbook on Ontologies*, pages 1–17. Springer, Berlin, Heidelberg, 2009. ISBN 978-3-540-92673-3. doi: 10.1007/978-3-540-92673-3_0.
- Mohamed Khemakhem, Cristina Valentini, Natascia Ralli, Sérgio Barros, Georg Löckinger, Federica Vezzani, Ana Salgado, Zhenling Zhang, Sabine Mahr, Sara Carvalho, Klaus Fleischmann, and Rute Costa. Terminology Management Meets AI: The ISO/TC 37/SC 3/WG 6 Initiative. In Katerina Gkirtzou, Slavko Žitnik, Jorge Gracia, Dagmar Gromann, Maria Pia di Buono, Johanna Monti, and Maxim Ionov, editors, *Proceedings of the 5th Conference on Language, Data and Knowledge: TermTrends 2025*, pages 16–24, Naples, Italy, September 2025. Unior Press. ISBN 978-88-6719-334-9.
- George Lakoff. *Women, Fire, and Dangerous Things: What Categories Reveal about the Mind*. University of Chicago Press, Chicago, IL, April 1990. ISBN 978-0-226-46804-4.
- Gregory L. Murphy. *The Big Book of Concepts*. The MIT Press, July 2002. ISBN 978-0-262-28035-8. doi: 10.7551/mitpress/1602.001.0001.
- Eleanor Rosch. Cognitive representations of semantic categories. *Journal of Experimental Psychology: General*, 104(3):192–233, 1975. ISSN 1939-2222. doi: 10.1037/0096-3445.104.3.192.
- Eleanor H. Rosch. Natural categories. *Cognitive Psychology*, 4(3):328–350, May 1973. ISSN 0010-0285. doi: 10.1016/0010-0285(73)90017-0.

Barry Smith. Beyond Concepts: Ontology as Reality Representation. In Barry Smith and Christopher Welty, editors, *Formal Ontology in Information Systems (FOIS)*, pages 1–12. ACM Press, 2001.

Rita Temmerman. *Towards New Ways of Terminology Description*. John Benjamins Publishing Company, 2000. ISBN 978-90-272-2326-5. doi: 10.1075/tlrp.3.